

# Infranet — Combined Formal Brief

Design · business feasibility · phased rollout — single canonical document

| Field          | Value                                                                                        |
|----------------|----------------------------------------------------------------------------------------------|
| Document type  | Combined formal brief (design + founder/business evaluation)                                 |
| Status         | Draft for human review — <b>not</b> a live product, fundraising deck, or implemented network |
| Date           | 2026-07-12 (combined / reconciled 2026-07-13)                                                |
| Canonical path | docs/infranet/INFRANET-COMBINED-BRIEF.md                                                     |
| PDF            | docs/infranet/INFRANET-COMBINED-BRIEF.pdf (when generated)                                   |
| R&D tree       | projects/infranet/ (demos + earlier architecture notes — <b>not deleted</b> )                |
| Project id     | infranet ( agents/user-tasks.json )                                                          |
| Audience       | Founder, advisors, multi-agent coordination                                                  |
| Supersedes     | INFRANET-DESIGN-PROPOSAL.md , INFRANET-BUSINESS-BRIEF.md (stubs only)                        |

## How to read labels

| Label      | Meaning                                                         |
|------------|-----------------------------------------------------------------|
| Decided    | Already true in this workspace / prior signed-off charter       |
| Proposed   | Design or policy choice for review; may change before any build |
| Assumption | Stated explicitly; change it and the conclusion may move        |
| Estimate   | Order-of-magnitude only; not audited market research            |
| Example    | Illustrative only — not a commitment to ship that product       |
| Demo-only  | From COST_ANALYSIS.md / in-repo spikes — not market GPU prices  |

## Document map

| Part     | Contents                                                           |
|----------|--------------------------------------------------------------------|
| Verdict  | Recommendation up front                                            |
| I        | How it works (design: tokens, marketplace, services, architecture) |
| II       | Business / market / unit economics / risks / rollout               |
| III      | Shared glossary, reconciled assumptions, recommendation checklist  |
| Appendix | Sources, reconciliation notes, COST_ANALYSIS caveat                |

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## Verdict (read this first)

**Recommendation: Proceed-with-conditions** on founder time and R&D — **not** yet on external capital or a public network launch.

Infranet's core thesis is sound as a *platform economics* story: monetize **coordination** (match, escrow, settlement, support) while **capacity stays on other people's idle machines**, and grow by recruiting supply rather than buying a GPU fleet. That is a real structural advantage versus cloud CAPEX — *if* attestation, quality, and cold-start demand can be solved for at least one narrow job class (**LLM inference** is the strongest first bet).

It is **not** yet investable as a scaled marketplace: demos are placeholders, idle-hardware quality is uneven, regulation around tokens/on-ramps is non-trivial, and chicken-egg supply/demand will dominate the first 12–24 months. Conditions to continue are in Part III.

Demos under `projects/infranet/` are research spikes. Homelab pieces (linuxbox Hermes, free-first model routing, resource governance) are **related practice**, not the product itself.

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## Part I — How Infranet works (design)

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### I.1 One-sentence pitch

**Infranet** is a settlement and marketplace layer whose unit of account is **attested compute** (and related network work: storage, verification, relay). Producers earn by completing verified jobs on **hardware they already own**; consumers and apps spend those claims to buy work—especially **LLM and agent jobs**—so useful capacity can circulate **without every hop requiring cash**. The platform takes a **thin skim** for coordination; optional fiat on/off-ramps sit beside the exchange, not inside every service call.

## I.2 Business model in four moves

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- |             |                                                                                                                                                                                              |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. SUPPLY   | People (and labs) contribute unused/underused CPU/GPU/storage overnight or off-peak → producers earn compute tokens.                                                                         |
| 2. DEMAND   | Users and apps buy work (esp. LLM/agent jobs) priced in tokens. Newcomers may buy tokens with cash (on-ramp); daily loop can stay token ↔ work without a card swipe each call.               |
| 3. APPS     | Independent services (social, tools, funding, meshes...) pay a modest entry fee to list on the registry, then settle SKUs in tokens.                                                         |
| 4. PLATFORM | Infranet Corp / founder operates coordination: matching, escrow, dispute, identity hooks, support, optional regulated on/off-ramps – funded by a small % of settled volume + app entry fees. |

**Idle-hardware thesis (Assumption / Proposed):** the company does **not** buy a large hardware fleet. Scaling = more idle capacity online + better matching. Company spend is software, ops, compliance, and thin always-on coordination (which can itself start on a homelab).

## I.3 Actors

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| Actor                      | What they do                                                    | How they pay / earn                                                      |
|----------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------|
| <b>Producer</b>            | Offers idle GPU/CPU/storage; runs jobs; posts attestation       | Earns work tokens (minus slash risk on fraud)                            |
| <b>Consumer</b>            | Buys inference, storage, agent runs, app features               | Spends tokens (or cash → tokens once)                                    |
| <b>App builder</b>         | Publishes a paid service (SKU + fulfillment + acceptance rules) | Pays <b>entry fee</b> ; settles in tokens; may take their own app margin |
| <b>Platform (Infranet)</b> | Match, escrow, dispute, registry, support, optional ramps       | Skim on volume + entry fees + (later) premium support tiers              |
| <b>Verifier (optional)</b> | Spot-checks / re-executes / attest rules                        | Small share of fees or protocol rewards                                  |

## I.4 Problem / motivation

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Useful digital work—especially inference, agents, verification, and coordination—is gated by:

1. **Cash-first markets.** Spare compute and demand rarely meet without a company in the middle taking a cut and setting terms.
2. **Opaque units.** Vendor “credits,” API keys, and platform points do not transfer cleanly.

3. **Services stuck to platforms.** Each app reinvent billing; no shared settlement that says *this payment buys this kind of attested work*.
4. **Trust without receipts.** Proving delivery usually means trusting a vendor dashboard.

This workspace already struggles productively with **resource governance** (search tokens, memory, free vs paid models, correctness). That experience is evidence of the *need*—not proof that Infranet already exists as a network.

| Prior material                               | What it covers                                                    | Status                              |
|----------------------------------------------|-------------------------------------------------------------------|-------------------------------------|
| projects/infranet/PLANNING.md / README.md    | Identity, black-box verification (FHE/MPC/ZKP sketch), NFC/mobile | Earlier architecture notes          |
| projects/infranet/COMPUTE_REWARDS.md         | Marketplace, rewards, reputation, network-internal tokenomics     | <b>Closest ancestor</b>             |
| projects/infranet/BLOCKCHAIN_PLATFORM.md     | IVM / InfranetScript sketch                                       | Sketch                              |
| projects/infranet/COST_ANALYSIS.md           | Gas/compute-unit cost models for crypto ops                       | <b>Demo-only</b> ≠ market GPU price |
| Python/Rust demos & tests                    | In-memory blockchain, identity, marketplace lifecycle             | <b>Placeholders</b> , not real FHE  |
| Potato BASELINE.md (SSH-verified 2026-07-12) | Source-trace of demo behavior                                     | Confirms no live crypto             |

## I.5 Vision and goals

You should be able to **earn** compute by offering capacity you already own, **spend** it on LLM/agents and other services, and **build new paid apps** on the same unit—without forcing a card swipe for every like, relay, or agent run—while the platform takes only a thin coordination skim.

| ID | Goal                                 | Success looks like                                                  |
|----|--------------------------------------|---------------------------------------------------------------------|
| G1 | <b>Crypto-defined compute tokens</b> | Issuance and spend tied to attested compute / verified network work |
| G2 | <b>Cashless trade</b>                | Consumer ↔ producer settle in tokens; fiat bridges optional         |
| G3 | <b>Service layer</b>                 | Apps (esp. LLM/agent) price and settle in the same unit             |
| G4 | <b>User-defined paid services</b>    | Publish a service that accepts/pays tokens under clear terms        |

**Non-goals:** replacing banks; guaranteeing free unlimited LLM; shipping a new L1 tomorrow; hard-coding only the example apps; treating linuxbox/Hermes as production Infranet.

### Design principles

1. **Work backs value.** Tokens track useful attested work—not vibes.
2. **Cash is a bridge, not the product.** Daily loop is token ↔ work.
3. **Receipts over promises.** Signed job specs, attested results, dispute rules.
4. **Ponytail growth.** Smallest working settlement first; FHE/identity ambitions stay available but are not MVP blockers.
5. **Honest limits.** Hardware, latency, and model quality vary; pricing must admit that.

**Decided** near-term honesty: ship the smallest settlement loop that moves a balance when a real or mocked inference job completes. Per `AGENT-CHARTER.md`, this remains R&D until explicitly promoted.

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## I.6 Pillar 1 — Crypto-defined compute tokens

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### Working definition (**Proposed**)

A **compute token** is a transferable claim that can be:

- **Earned** by completing attested jobs (or by protocol-approved issuance tied to staking / capacity proofs), and
- **Spent** to purchase jobs or service features that settle on the same ledger.

| It is                                                  | It is not                              |
|--------------------------------------------------------|----------------------------------------|
| A ledger claim usable to buy attested work             | A memecoin or “number go up” product   |
| Earned primarily by doing (or escrowing for) real jobs | A promise of free unlimited LLM access |
| Bridged to cash at the edge when users need it         | Mandatory cash on every micro-action   |

### What cryptography is for

| Job                      | Why crypto / ledger                                        |
|--------------------------|------------------------------------------------------------|
| <b>Minting rules</b>     | Who can create tokens, under what proof                    |
| <b>Ownership</b>         | Who can spend which balance                                |
| <b>Escrow</b>            | Lock payment until delivery conditions are met             |
| <b>Auditability</b>      | Settlement history without trusting one company’s database |
| <b>Attestation hooks</b> | Tie release of funds to evidence of work                   |

### Issuance, mint, burn (**Proposed — reconciled**)

Two issuance modes can coexist; Phase 1 picks one primary story:

1. **Escrow-transfer model (preferred for marketplace):** Consumer locks tokens → producer delivers attested work → escrow releases. Net supply unchanged; tokens move

consumer → producer (minus platform skim).

2. **Work-mint model (protocol bootstrap):** Protocol mints a capped amount to producers on attested *network-internal* work (verification, consensus, storage integrity)—spirit of `COMPUTE_REWARDS.md` §3–§6.

**Burn / sinks:** small **platform skim** on settled jobs (see Part II / reconciled fee table); optional burn on spam or slash events. No hidden inflation.

**Reconciled:** the historical `COMPUTE_REWARDS.md` §6.1 split (**40/30/15/10/5**) applies only to *network-internal* reward sketches — **not** locked for the service marketplace or corporate P&L.

### Token classes (**Proposed; Phase 1 may collapse**)

| Class                              | Backing / issuance                                           | Use                                          |
|------------------------------------|--------------------------------------------------------------|----------------------------------------------|
| <b>Work tokens (primary)</b>       | Minted/unlocked on attested completion, or escrow → producer | Day-to-day pay for inference, storage, relay |
| <b>Stake / bond</b>                | Locked by producers (and maybe services)                     | Slash on fraud; matching priority            |
| <b>Service vouchers (optional)</b> | Issued by a service for prepaid packages                     | UX sugar redeemable into work tokens or jobs |

**Proposed MVP:** one fungible work token + simple stake locks.

### Measuring “compute” without fake precision (**Proposed**)

Do **not** pretend one universal FLOP unit prices every job fairly. Prefer:

1. **Job-class meters** — LLM: input/output tokens × model id × hardware tier; storage: GB-month; relay: packet-bytes × uptime.
2. **Posted prices** — producers name a rate; market clears.
3. **Attestation suited to the class** — signed result hash, TEE, random re-execution, ZK where it pays.

### Wallets and cash bridges (**Proposed**)

- Ledger accounts (keypair-backed) for balances and escrow.
  - Services may hold treasury / payout accounts.
  - UX may hide chain jargon (“compute credits”) while the ledger records ownership.
  - **Decided:** no secrets, private keys, or seed phrases in repo docs or git.
  - **On-ramp / off-ramp:** optional; regulated entity or peer OTC — policy choice. Default story: earn overnight → spend daytime → cash only when someone wants fiat.
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## I.7 Pillar 2 — Trading without mandatory cash

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### Atomic transaction (happy path)

1. Producer stakes a bond and advertises capacity (model tier, latency, privacy).
2. Consumer or app **escrows** work tokens against a job spec.
3. Producer executes; returns result + attestation.
4. Escrow **releases** to producer; platform skim and optional verifier fee peel off.
5. Producer spends tokens on other apps / agent tools — **cash optional**.

Same pattern for every vertical: only the **job class** and **acceptance rules** change.

### Job as the atomic object (Proposed)

Offer → Request → Match → Escrow → Execute → Settle → Dispute (short window; reputation updates).

### Matching modes (Proposed)

| Mode                      | When                                                  |
|---------------------------|-------------------------------------------------------|
| Spot market               | One-off jobs; price discovery                         |
| Subscriptions / retainers | Locked tokens for recurring capacity                  |
| Pools                     | Many small producers behind one gateway (LLM pattern) |
| Direct peer (P2P)         | Known counterparties; marketplace only settles        |

### Reputation, privacy, cashless peer exchange (Proposed)

- Reputation from completed jobs and dispute outcomes; bonds that slash on fraud.
- Privacy tiers: Public → Encrypted transit → Confidential compute (TEE/MPC when priced in). LLM prompts usually need at least encrypted transit; full FHE for every inference is the wrong first bet.
- Work ↔ work, OTC, and service-mediated flows need **no** fiat conversion API per trade.

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## I.8 Pillar 3 — Services on the token (especially LLM)

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### First product: LLM / agent gateway (Proposed)

**Why first:** Clear unit of work, huge demand, natural fit for spare GPU, dogfoodable on this stack.

**Gateway responsibilities:** accept payment in work tokens; translate to job specs; route to producers; return attested results; expose human chat and agent/tool APIs.

**What it is not (initially):** a promise to beat every cloud provider on price, or a single global model monopoly.

**Analogy (not product):** Hermes free-first → paid fallback is a *routing policy*. Infranet would turn “who gets paid for which inference” into a **settled job** with a token balance.

| Element      | Detail                                                      |
|--------------|-------------------------------------------------------------|
| SKU examples | Chat completion, embed batch, agent tool loop (max N steps) |
| Meter        | Tokens in/out × model id × tier                             |
| Producer     | Home GPU daemon, lab box, or pool operator                  |
| Settlement   | Escrow → attest → release                                   |
| Failover     | Next producer on SLA miss; partial refund rules per SKU     |

Any service that defines a **SKU**, a **fulfillment job**, and **acceptance rules** can settle on Infranet.

**Adjacent examples:** encrypted storage (GB-month); verification labor; relay/bandwidth for mesh apps.

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## I.9 Pillar 4 — User-defined paid services

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Illustrative verticals—**Examples**, not a commitment to build all of them.

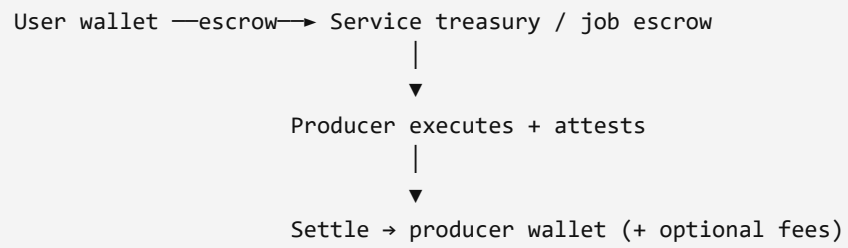
| Vertical                | Pay tokens for                                        | Earn by                             | Notes                                                        |
|-------------------------|-------------------------------------------------------|-------------------------------------|--------------------------------------------------------------|
| Decentralized social    | Boosts, hosting, media store, mod labor, ranking runs | Store/relay/moderate                | Cashless tips → hosting                                      |
| Road / short-range mesh | Priority relay, authenticated alerts, map layers      | Radios/edge boxes, attested sensors | <b>High-risk</b> — legal/safety before any public-road pilot |
| Todo / agent tools      | Recurring agent runs, private sync                    | Scheduled jobs + encrypted storage  | Earn night / spend day                                       |
| Arts crowdfunding       | Milestone locks                                       | Render, distribution, outreach      | Escrow on attested delivery                                  |

### Service registry (Proposed)

- Manifest: name, SKU schema, job-class ids, payment addresses, policy URLs
- Versioned interfaces so new verticals do not fork the token
- **Paid entry fee** (cheap, not free-subsidized)—filters spam; early platform oxygen



- **Reconciled timing:** charge entry fees only after the LLM gateway has real external usage (Phase 3) — do not subsidize a ghost registry (see Part III conditions)



## I.10 Architecture sketch

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```
flowchart TB
    subgraph apps [Service apps]
        LLM[LLM / agent gateway]
        Social[Social / media]
        Road[Road / mesh relay]
        Todo[Todo / agents]
        Arts[Arts escrow]
        More[Future services...]
    end

    subgraph market [Exchange and marketplace]
        Match[Match / pools]
        Escrow[Escrow]
        Dispute[Dispute / reputation]
    end

    subgraph token [Token and attestation]
        WorkTok[Work tokens]
        Stake[Stake / bonds]
        Attest[Attestation rules]
    end

    subgraph nodes [Producer capacity]
        GPU[GPU / CPU inference]
        Store[Storage]
        Radio[Short-range radios]
        Verify[Verifiers / attestors]
    end

    Cash[Optional fiat on/off-ramp]

    apps --> Match
    Match --> Escrow
    Escrow --> WorkTok
    Escrow --> Attest
    Attest --> nodes
    nodes --> Escrow
    Stake --> Match
    Cash -. -> WorkTok
    Dispute --> Stake
    Dispute --> WorkTok
```

| Layer                       | Prior docs                       | This brief                                                        |
|-----------------------------|----------------------------------|-------------------------------------------------------------------|
| Identity / black-box verify | PLANNING, RESEARCH               | Optional strength for high-trust jobs                             |
| Programmable substrate      | BLOCKCHAIN_PLATFORM (IVM sketch) | One substrate option among several ( <b>open question</b> )       |
| Network-internal rewards    | COMPUTE_REWARDS                  | Feeds issuance/reputation; marketplace payment is consumer escrow |
| Service economy             | ( <i>this doc</i> )              | North star for <i>why</i> tokens exist                            |

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## I.11 Trust, security, and abuse (Proposed)

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| Threat                        | Direction                                                              |
|-------------------------------|------------------------------------------------------------------------|
| Fake work / garbage inference | Bonds, spot re-execution, allowlists, reputation                       |
| Spam job posts                | Posting deposits / burns; rate limits                                  |
| Sybil producers               | Stake, identity for high-trust tiers, diversity-aware matching         |
| Griefing disputes             | Short windows; bond to challenge; clear evidence rules                 |
| Speculation-first token       | Tie primary issuance to jobs; no yield theater in core                 |
| Double-spend                  | Single ledger; escrow locks before work; settlement is source of truth |
| Centralization of supply      | Diversity-aware matching; small-producer pools; honest UX              |
| Mobility / surveillance       | Delay public-road pilots; jurisdiction gates; no evasion tooling       |

Carry forward abuse list from `COMPUTE_REWARDS.md` §8 (Sybil, collusion, free-riding, result manipulation, resource inflation).

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## I.12 Open design questions

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1. **Substrate:** custom chain vs public chain vs thin application ledger for MVP?
2. **One token or many** at Phase 1?
3. **Attestation minimum** for paid LLM jobs (signature only vs TEE vs redundant execution)?
4. **Who runs cash on/off-ramps**, and where?
5. **Governance:** who changes fees and job-class rules?
6. **Default privacy** for prompts and for location/mobility data?
7. **UX naming:** “compute credits” vs on-chain jargon?
8. **Scope merge:** does compute-as-currency supersede identity-first README marketing, or stay parallel? (`user-tasks.json` asks for charter reconcile.)

9. **First spike owner:** PC lab vs potato always-on vs both?

# Part II — Business, market, feasibility, and rollout

## II.1 Market opportunity

### Problem the market already feels

- 1. Spare capacity is wasted (consumer/prosumer GPUs idle nights/weekends).
- 2. Demand for inference is surging — cash-gated via cloud vendors and API aggregators.
- 3. Billing is opaque and non-portable.
- 4. Building a paid app means reinventing payment + infra every time.

Infranet attacks (3)+(4) with shared settlement, and (1)+(2) by matching idle supply to inference demand without company-owned fleets.

### Market framing (Estimate / Assumption)

Figures are order-of-magnitude planning bands — not a commissioned TAM study.

| Layer                            | Scope                                                                                  | Estimate band                                                                                                  | Notes                                                               |
|----------------------------------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| <b>TAM (loose)</b>               | Global cloud + GPU rental + decentralized compute + LLM API spend                      | <b>High tens of billions USD/year</b>                                                                          | Ceiling only                                                        |
| <b>SAM</b>                       | Spot / flexible GPU inference + peer compute markets + indie LLM gateway spend         | <b>Low-mid single-digit billions USD/year</b>                                                                  | Overlaps Render/Akash/Golem/cloud spot and OpenRouter-like gateways |
| <b>SOM (5-year, conditional)</b> | Attested idle-GPU inference + a handful of token-settled apps in reachable communities | <b>\$1M–\$30M/year GMV</b> settled; platform revenue = skim + entry fees → likely <b>well under \$1M</b> early | Cold-start capped; not a Series-A fantasy                           |

**Assumption:** SOM assumes successful Phase 0–2 and **does not** assume winning against AWS/GCP on enterprise SLA.

## II.2 Competitive landscape

| Player / pattern                          | What they optimize           | Overlap                  | Differentiation to claim                                                              |
|-------------------------------------------|------------------------------|--------------------------|---------------------------------------------------------------------------------------|
| Hyperscale cloud GPU                      | CapEx fleets, enterprise SLA | Reliable inference       | No fleet CAPEX; cashless micro-settlement across apps                                 |
| GPU rental / render nets                  | Artists & batch GPU          | Idle GPUs                | General <b>job escrow + multi-app currency</b>                                        |
| Decentralized compute (Akash, Golem, ...) | Container/VM markets         | Closest cousins          | <b>Service registry + user-defined paid apps</b> ; LLM gateway as first product story |
| Storage / data markets                    | Storage proofs               | Adjacent capacity        | Later job class; not the first wedge                                                  |
| LLM gateways (OpenRouter-style)           | Model routing, API UX        | Demand-side UX           | Portable compute claims; producers can be homes                                       |
| Points / credits inside SaaS              | Lock-in billing              | Psychological competitor | Portability + open app registry                                                       |

**Positioning (Proposed):** Infranet is not “cheaper AWS.” It is a **thin settlement OS for attested work**, with idle hardware as the supply engine and paid apps as the demand engine.

**Platform test:** Infranet is a *platform* when a third-party app can (1) pay entry fee, (2) define SKUs + job classes, (3) settle without forking the token, (4) rely on shared producers. Until (4) holds for  $\geq 2$  apps, call it a **product with platform ambition**. If only one LLM gateway ever works, the platform thesis fails (narrow product may still be worth pursuing under a smaller ambition).

## II.3 Value proposition by stakeholder

| Stakeholder         | Value                                                                                             |
|---------------------|---------------------------------------------------------------------------------------------------|
| <b>Producers</b>    | Monetize machines already bought; overnight earnings; spend in-ecosystem without cashing out      |
| <b>Consumers</b>    | Potentially cheaper/flexible inference; portable balance across apps; cash optional after on-ramp |
| <b>App builders</b> | Skip reinventing wallets/escrow; paid entry keeps spam down; SKUs in one currency                 |
| <b>Platform</b>     | Asset-light growth; revenue from skim + entry fees; alignment with volume rather than owning GPUs |

## II.4 Unit economics (**Proposed** — reconciled)

### Reconciled fee / token policy (single source of truth)

| Parameter                                                           | Working band                                                                               | Status                       | Rationale                                                       |
|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------|------------------------------|-----------------------------------------------------------------|
| <b>Platform skim</b> on settled job volume                          | <b>0.05%–0.5%</b>                                                                          | <b>Proposed</b>              | Infrastructure cut; stay below painful 15–30% marketplace rents |
| Working point for illustrations                                     | <b>~0.2%</b>                                                                               | <b>Example</b>               | Used only for sensitivity math below                            |
| Verifier / dispute share                                            | Split from skim or tiny add-on                                                             | <b>Proposed</b>              | Pay for integrity without bloating price                        |
| App registry <b>entry fee</b>                                       | <b>Paid, not free</b> — <b>Example \$50–\$500</b> one-time or annual (or token-equivalent) | <b>Proposed</b>              | Filters spam; early cash oxygen                                 |
| When entry fees start                                               | <b>Phase 3</b> — after gateway has real external usage                                     | <b>Proposed / reconciled</b> | Do not subsidize a ghost registry                               |
| Network-internal reward split                                       | 40/30/15/10/5 in <code>COMPUTE_REWARDS.md</code>                                           | <b>Sketch only</b>           | Not corporate P&L; not service-marketplace fees                 |
| Token ↔ compute pricing                                             | Job-class meters + posted rates                                                            | <b>Proposed</b>              | No fake universal FLOPs                                         |
| <code>COST_ANALYSIS.md</code> (1 token ≈ \$0.01, tasks ~\$0.10–\$1) | Demo ledger arithmetic                                                                     | <b>Demo-only</b>             | Not market GPU-hour or revenue forecast                         |

#### Illustrative math (Assumption):

If annual settled GMV = \$10M and skim = 0.2%, platform take ≈ **\$20k/year** — too thin alone. Early viability **depends on app entry fees + focused gateway usage**, not skim. Skim becomes meaningful only at much larger GMV (**Assumption:** \$500M+ GMV × 0.2% ≈ \$1M — far beyond SOM without a breakout).

**Implication (reconciled):** treat skim as **alignment + long-term upside**; treat **entry fees + optional premium coordination** as **near-term oxygen**.

#### When cash enters

1. App entry fee → corporate treasury (cash or prepaid tokens).
2. Consumer on-ramp → tokens under policy.
3. Producer off-ramp → tokens → fiat (compliance-gated).
4. Volume skim → mostly tokens initially; convert for ops.

5. Day-to-day job settle → **no** cash (token escrow → release).

## Company cost structure

| Cost                                                  | Nature                                                          |
|-------------------------------------------------------|-----------------------------------------------------------------|
| Software (ledger, gateway, producer daemon, registry) | Primary build cost                                              |
| Always-on coordination hosts                          | Small vs GPU fleets; can start on existing linuxbox/PC          |
| Support, dispute ops                                  | Scales with users                                               |
| Compliance / legal (ramps, mobility)                  | Spiky; do not defer forever                                     |
| Marketing                                             | Needed for cold start; avoid burning cash on free app subsidies |

**Non-cost (by design):** owning and refreshing a large GPU fleet.

## Sensitivity

| If this is wrong...                        | Then...                                                                          |
|--------------------------------------------|----------------------------------------------------------------------------------|
| Idle GPUs cannot meet quality/SLA          | Gateway must blend pools + cloud fallback → margin compresses; thesis weakens    |
| Users refuse any token UX                  | Product becomes "credits" UI over ledger — still OK if settlement stays portable |
| Skim stays tiny and apps refuse entry fees | No business — park or become pure open-source protocol                           |
| Only LLM gateway works                     | Narrow company, not platform — may still be worth a focused product              |

## II.5 Risks and mitigations

| Risk                            | Why it matters                         | Mitigation direction                                                                          |
|---------------------------------|----------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Cold start / chicken-egg</b> | No producers ↔ no jobs                 | Closed start: friends' GPUs + one paid demand source; paid app entry only after gateway works |
| <b>Quality of idle hardware</b> | Throttling, flaky uptime, malware      | Tiers, bonds, reputation, allowlists, spot re-execution; honest UX (not "enterprise SLA")     |
| <b>Fraud / fake work</b>        | Garbage inference                      | Escrow + attestation ladder; slash stakes                                                     |
| <b>Trust &amp; privacy</b>      | Prompts on strangers' PCs              | Privacy tiers; encrypted transit default for LLM                                              |
| <b>Regulation</b>               | Token as security?, money transmission | Delay public ramps; legal review before public token sales                                    |
| <b>Centralization of supply</b> | Farms capture matching                 | Diversity-aware matching; transparent capture metrics                                         |
| <b>Speculation capture</b>      | Token detaches from work               | Issuance tied to jobs/escrow; no yield theater                                                |
| <b>Road/mobility vertical</b>   | Legal/safety landmine                  | Explicitly late; policy review before public-road pilot                                       |
| <b>Founder bandwidth</b>        | Competing live stack                   | Phase gates; one spike owner; R&D charter                                                     |

## II.6 Vertical priority

| Priority     | Vertical                               | Why / why not                                                  |
|--------------|----------------------------------------|----------------------------------------------------------------|
| <b>P0</b>    | LLM / agent gateway                    | Clear meter, huge demand, fits idle GPU, dogfoodable           |
| <b>P1</b>    | Agent-powered todos / household tools  | Earn night / spend day; lower regulatory heat                  |
| <b>P1</b>    | Arts crowdfunding + render             | Escrow milestones + GPU; shared settlement vs Render-only      |
| <b>P2</b>    | Decentralized social                   | Network effects if settlement sticks; moderation hard          |
| <b>P3</b>    | Encrypted storage / verification labor | Complements identity track; slower consumer story              |
| <b>Defer</b> | Road / short-range awareness mesh      | High legal/safety risk — architecture-compatible, product-late |



## II.7 Phased roadmap (**Proposed** — reconciled design + GTM)

Dates are **order of work**, not calendar promises. Ponytail: smallest correct slice first.

| Phase | Name                               | Goals                                                                                                                   | Verify                                                            | GTM                                                                                                  |
|-------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| 0     | Homelab truth (now)                | Smallest settlement loop; optional meter one OpenRouter/local call against mock ledger; align this brief                | One runnable check in <code>projects/infranet/</code>             | None public — founder + agents                                                                       |
| 1     | Friends & family supply            | Work token + escrow on chosen substrate (thin ledger OK); LLM job class only; manual cash on-ramp for testers           | 3–10 producer nodes; signed results                               | Invite-only<br>Discord/Tailscale                                                                     |
| 2     | Niche LLM gateway                  | HTTP/API gateway priced in tokens; installable producer daemon; attestation v1 + failure/refund rules                   | External test users finish real chats/agent loops                 | One niche (agent builders / dogfood); compete on portable settlement + idle supply — not GPT latency |
| 3     | First non-LLM app + paid entry     | Pick <b>one</b> : agent-todo, arts escrow, or social storage/boost; charge entry fee; prove earn-night / spend-day path | Registry/SKU works; fee received; ≥1 real cashless cross-app path | Builders wanting paid features without Stripe-per-micro-action; <b>keep mobility out</b>             |
| 4     | Harder verticals / stronger crypto | Identity/black-box where trust pays; multi-class tokens only if needed; mobility only after legal review                | Policy gates cleared                                              | Cautious                                                                                             |
| 5     | Scale & governance                 | Fee finalization, treasury rules, external audit, explicit production promotion per <code>AGENT-CHARTER.md</code>       | Audit + promotion decision                                        | Public cautious open                                                                                 |

Homelab spike → Friends' GPUs → Niche LLM gateway → Paid app #2 → Public cautious open  
↑ \_\_\_\_\_ supply side growth (idle capacity) \_\_\_\_\_ |

**Homelab vs long-term:** Phases 0–2 can live on PC + linuxbox without claiming a public network. Phases 3–5 need deliberate promotion.

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## Part III — Glossary, assumptions, recommendation

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### III.1 Shared glossary

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| Term                       | Meaning in this brief                                                     |
|----------------------------|---------------------------------------------------------------------------|
| <b>Work token</b>          | Fungible ledger claim used to buy attested jobs / service SKUs            |
| <b>Escrow</b>              | Tokens locked until acceptance rules say the job is done                  |
| <b>Attestation</b>         | Evidence of work (signature, TEE quote, re-execution, ZK where justified) |
| <b>Platform skim</b>       | Tiny % of settled volume paid to Infranet for coordination                |
| <b>Entry fee</b>           | Paid cost for an app to list on the service registry                      |
| <b>Job class</b>           | Category of work with its own meter, attestation, and dispute rules       |
| <b>SKU</b>                 | User-facing purchasable feature that creates one or more jobs             |
| <b>Producer / consumer</b> | Supply node vs demand side (user or app)                                  |
| <b>On/off-ramp</b>         | Optional fiat ↔ token bridge at the edge                                  |
| <b>Substrate</b>           | Where the ledger lives (custom chain, public chain, or thin app ledger)   |

## III.2 Reconciled assumptions (do not fork these)

| #   | Assumption                                                                                            | Source of truth        |
|-----|-------------------------------------------------------------------------------------------------------|------------------------|
| A1  | Company does <b>not</b> CAPEX a large GPU fleet; growth = idle supply online                          | Parts I–II             |
| A2  | Skim band <b>0.05%–0.5%</b> of settled job volume (illustrative mid <b>~0.2%</b> )                    | Part II.4              |
| A3  | Near-term oxygen = <b>entry fees + gateway usage</b> ; skim is long-term alignment                    | Part II.4              |
| A4  | Entry fee <b>Example \$50–\$500</b> ; charged from <b>Phase 3</b> after real gateway usage            | Part II.4 / II.7       |
| A5  | MVP = <b>one work token</b> + simple stakes; escrow-transfer preferred over work-mint for marketplace | Part I.6               |
| A6  | COMPUTE_REWARDS 40/30/15/10/5 and COST_ANALYSIS \$0.01/token math are <b>not</b> market or P&L policy | Part II.4 / Appendix B |
| A7  | First wedge = <b>LLM / agent gateway</b> ; mobility deferred                                          | Part II.6              |
| A8  | Demos / Hermes / resource-governance practice $\neq$ live Infranet product                            | Verdict / Part I.4     |
| A9  | TAM/SAM/SOM figures are <b>Estimate</b> bands for founder judgment                                    | Part II.1              |
| A10 | Production requires <b>explicit</b> promotion per AGENT-CHARTER.md                                    | Part II.7 Phase 5      |

## III.3 Recommendation

### Decision

| Option                         | Meaning                                | Choice       |
|--------------------------------|----------------------------------------|--------------|
| <b>Proceed</b>                 | Full public build / raise now          | No           |
| <b>Proceed-with-conditions</b> | Continue R&D and Phase 0–2 under gates | <b>Yes</b>   |
| <b>Park</b>                    | Stop until market/timing changes       | No (not yet) |

### Conditions to keep going

1. **Ship Phase 0 settlement spike** with one runnable check (balance moves on job complete).
2. **Pick substrate** for Phase 1 (thin app ledger vs public chain) — decide explicitly; do not boil the ocean on IVM first.
3. **LLM-only** until escrow+attestation hold under friend-group load.
4. **No public token sale / open on-ramp** until legal review.
5. **Paid app entry** only after gateway has real external usage — do not subsidize a ghost registry.
6. **Time-box founder hours** against higher-priority live stack work; Infranet stays charter R&D until promoted.

7. Revisit go/no-go after Phase 2 with metrics: jobs settled, producer uptime distribution, dispute rate, willingness to pay entry fee.

### Would you invest time?

**Yes — bounded time.** The economics story (idle supply + thin coordination skim + paid apps on shared settlement) is a coherent platform bet and matches how this stack already thinks about metered compute. Worth a **ponytail R&D track** through Phase 0–1 and a harsh Phase 2 review.

**Not yet — unbounded time or capital.** Without quality attestation and a real demand wedge, this becomes another underused decentralized compute paper.

### Human review checklist

- ☐ Core idea (compute as tradable attested work) — agree / revise
- ☐ Business model (idle supply + skim + entry fees) — agree / revise rates
- ☐ Cashless consumer ↔ producer loop — agree / revise
- ☐ LLM gateway as first wedge — agree / change
- ☐ Example verticals (social, mobility, todos, arts) — keep / drop / re-prioritize
- ☐ SOM honesty / competitive framing — agree / revise
- ☐ Phase 0–1 spike shape — agree / propose different smallest slice
- ☐ Proceed-with-conditions — accept / park / proceed harder
- ☐ Phase 0 spike owner (PC vs potato vs both)

Feedback: `AI_GROUPCHAT.md` ledger lines, or project `infranet` in `agents/user-tasks.json`.

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## Appendix A — Source map

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| Path                                                         | Role                                                      |
|--------------------------------------------------------------|-----------------------------------------------------------|
| docs/infranet/INFRANET-COMBINED-BRIEF.md                     | <b>This document — canonical</b>                          |
| docs/infranet/INFRANET-DESIGN-PROPOSAL.md                    | Stub → combined                                           |
| docs/infranet/INFRANET-BUSINESS-BRIEF.md                     | Stub → combined                                           |
| docs/infranet-proposal.md ,<br>projects/infranet/PROPOSAL.md | Stubs → combined                                          |
| projects/infranet/COMPUTE_REWARDS.md                         | Marketplace, reputation, historical token split           |
| projects/infranet/COST_ANALYSIS.md                           | Demo gas/token cost shapes ( <b>Demo-only</b> )           |
| projects/infranet/PLANNING.md                                | Identity / FHE-MPC-ZKP ambition (optional strength)       |
| projects/infranet/BLOCKCHAIN_PLATFORM.md                     | Substrate sketch (open question)                          |
| projects/infranet/AGENT-CHARTER.md                           | R&D vs production promotion                               |
| projects/infranet/README.md                                  | Project overview (identity-first heritage + pointer here) |
| .cursor/rules/resource-governance.mdc                        | Related practice — not the product                        |

Research under `projects/infranet/` is **retained**; this brief does not delete it.

### Evidence notes (2026-07-12 research pass)

- PC `projects/infranet/` holds the full R&D tree; Desktop sibling `MAIN_PROGRAMMING_FILES/Infranet/` was transferred 2026-06-28.
- Potato SSH confirmed `BASELINE.md`: FHE/MPC/ZKP are type labels in Python demos, not implementations.
- No laptop-kit Infranet product brief found in searchable paths that session; do not invent laptop content.

## Appendix B — COST\_ANALYSIS caveat (explicit)

`COST_ANALYSIS.md` examples (identity registration  $\approx$  \$0.02, simple transfer  $\approx$  \$0.21, FHE task rewards  $\approx$  \$0.10–\$1.00 under **1 token = \$0.01**) describe **demo ledger economics**, not:

- market price of an NVIDIA GPU-hour,
- OpenRouter token prices, or
- Infranet Corp revenue forecasts.

Use them to sanity-check that **protocol operations can be cheap relative to useful work** if the substrate cooperates — then price marketplace jobs from **posted producer rates**.

## Appendix C — Reconciliation log (design ↔ business)

| Topic                 | Design proposal said                      | Business brief said                          | Combined resolution                         |
|-----------------------|-------------------------------------------|----------------------------------------------|---------------------------------------------|
| Platform skim         | 0.05%–0.5%                                | 0.05%–0.5%                                   | <b>Identical</b> — kept                     |
| Entry fee amount      | “Cheap, not free”                         | Example \$50–\$500                           | Adopt business band as <b>Example</b>       |
| Entry fee timing      | Implied at registry launch                | Phase 3 after gateway usage; condition G.2.5 | <b>Business timing wins</b>                 |
| Idle hardware / CAPEX | No fleet; supply-side growth              | Same                                         | <b>Identical</b>                            |
| Token MVP             | One work token + stakes; escrow preferred | Same                                         | <b>Identical</b>                            |
| COMPUTE_REWARDS split | Historical sketch, not locked             | Sketch only, not P&L                         | <b>Identical</b> — labeled Demo/sketch      |
| Roadmap phases        | 0–5 design slices                         | 0–5 with GTM verify columns                  | <b>Merged</b> into one phase table (II.7)   |
| Recommendation        | (design had review checkboxes only)       | Proceed-with-conditions                      | <b>Business recommendation</b> is canonical |
| First product         | LLM gateway                               | LLM gateway P0                               | <b>Identical</b>                            |
| Mobility              | Example vertical, high-risk late          | Defer                                        | <b>Defer</b> until legal/safety review      |

## Document control

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| Field                 | Value                                                                |
|-----------------------|----------------------------------------------------------------------|
| Title                 | Infranet — Combined Formal Brief                                     |
| Canonical path        | docs/infranet/INFRANET-COMBINED-BRIEF.md                             |
| Supersedes            | INFRANET-DESIGN-PROPOSAL.md , INFRANET-BUSINESS-BRIEF.md (now stubs) |
| Implementation status | <b>Proposal / brief only</b>                                         |
| Recommendation        | <b>Proceed-with-conditions</b>                                       |